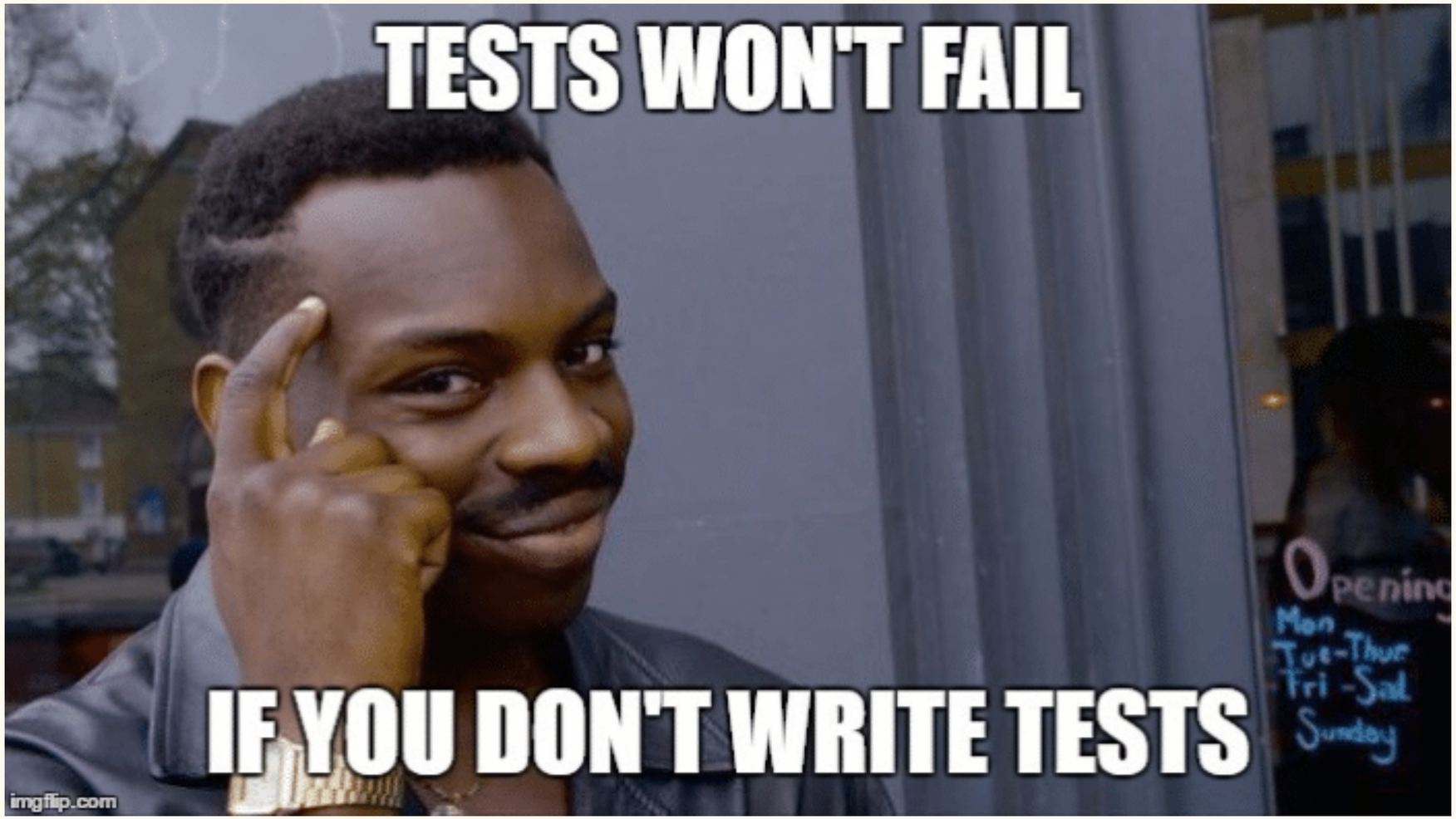
TESTING YOUR CODE


I DON'T ALWAYS TEST MY CODE
BUT WHEN I DO, I DO IT IN PRODUCTION


TESTS WON'T FAIL
IF YOU DON'T WRITE TESTS
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I. Lesson

A simple test module

```rust
#[cfg(test)]
mod tests {
    #[test]
    fn it_works() {
        assert_eq!("hello world!", "hello world!");
    }
}
```

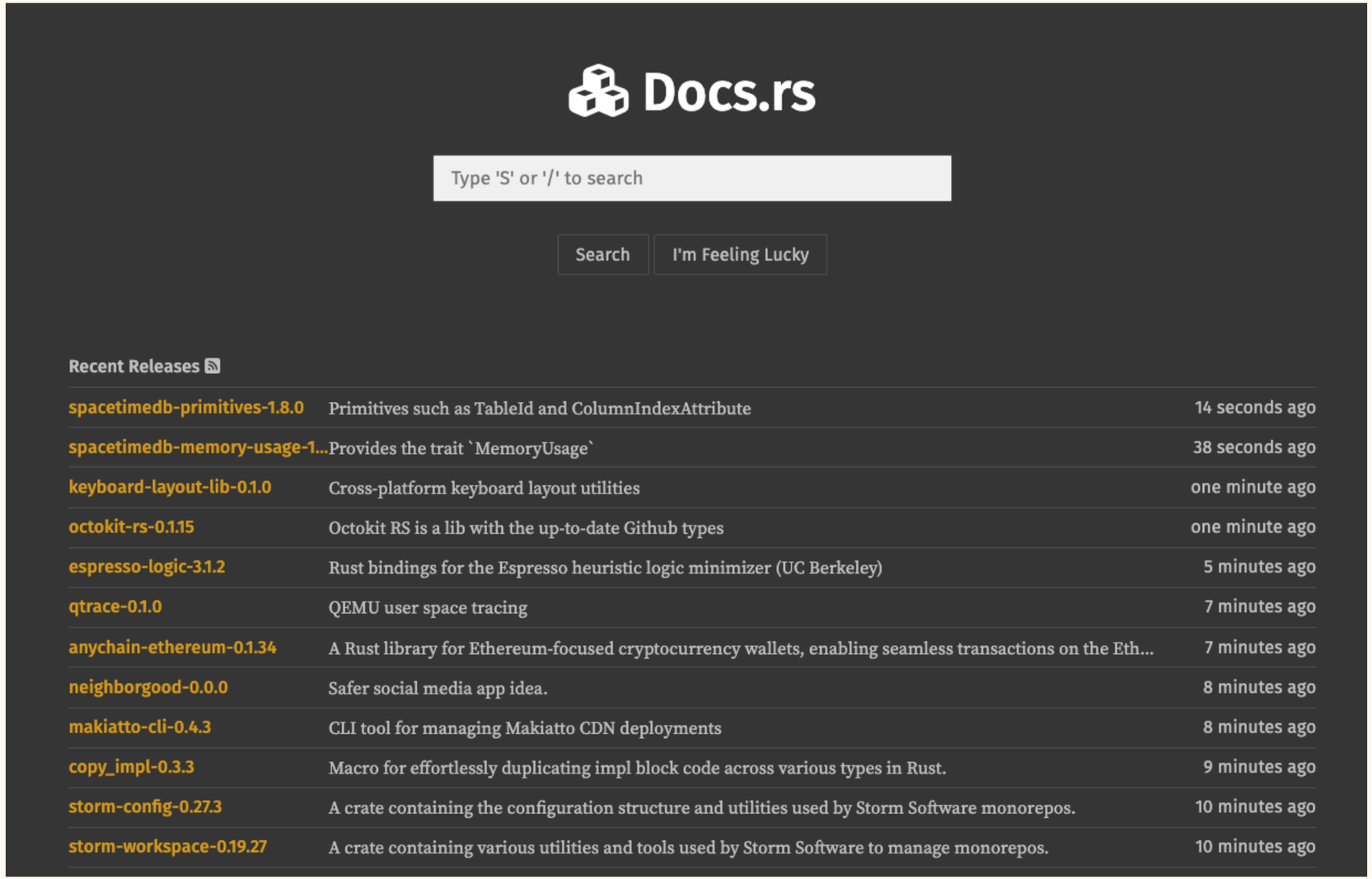
How cargo test works

How `cargo test` Works

1. **Cargo reads project metadata**
   - Reads `Cargo.toml` and dependencies.
2. **Compile library code**
   - Compiles `src/lib.rs` and dependencies.
3. **Compile test code**
   - **Unit Tests:** Inside `#[cfg(test)]` modules in library/binary files.
   - **Integration Tests:** Each file in `tests/` is compiled as a separate executable.
4. **Link test binaries**
   - Test code is linked with the compiled library.
5. **Run tests**
   - Executes all test binaries.
   - Reports passed/failed results.

Understanding Test Output

docs.rs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II. Project

Example with divide function

```rust
fn divide(a: i32, b: i32) -> Result<i32, String> {
    if b == 0 {
        return Err("Cannot divide by zero".to_string());
    }
    Ok(a / b)
}
```

└> We need to make sure this works!

```rust
#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn test_valid_division() {
        let result = divide(10, 2);
        assert_eq!(result, Ok(5));
    }

    #[test]
    fn test_division_by_zero() {
        let result = divide(10, 0);
        assert!(result.is_err());
    }

    #[test]
    fn test_negative_numbers() {
        let result = divide(-10, 2);
        assert_eq!(result, Ok(-5));
    }
}
```

Test panicking

```rust
fn get_item(index: usize) -> i32 {
    let items = vec![10, 20, 30];
    items[index]  // Panics if index out of bounds
}
```

└> *Will it panic properly?* 😤

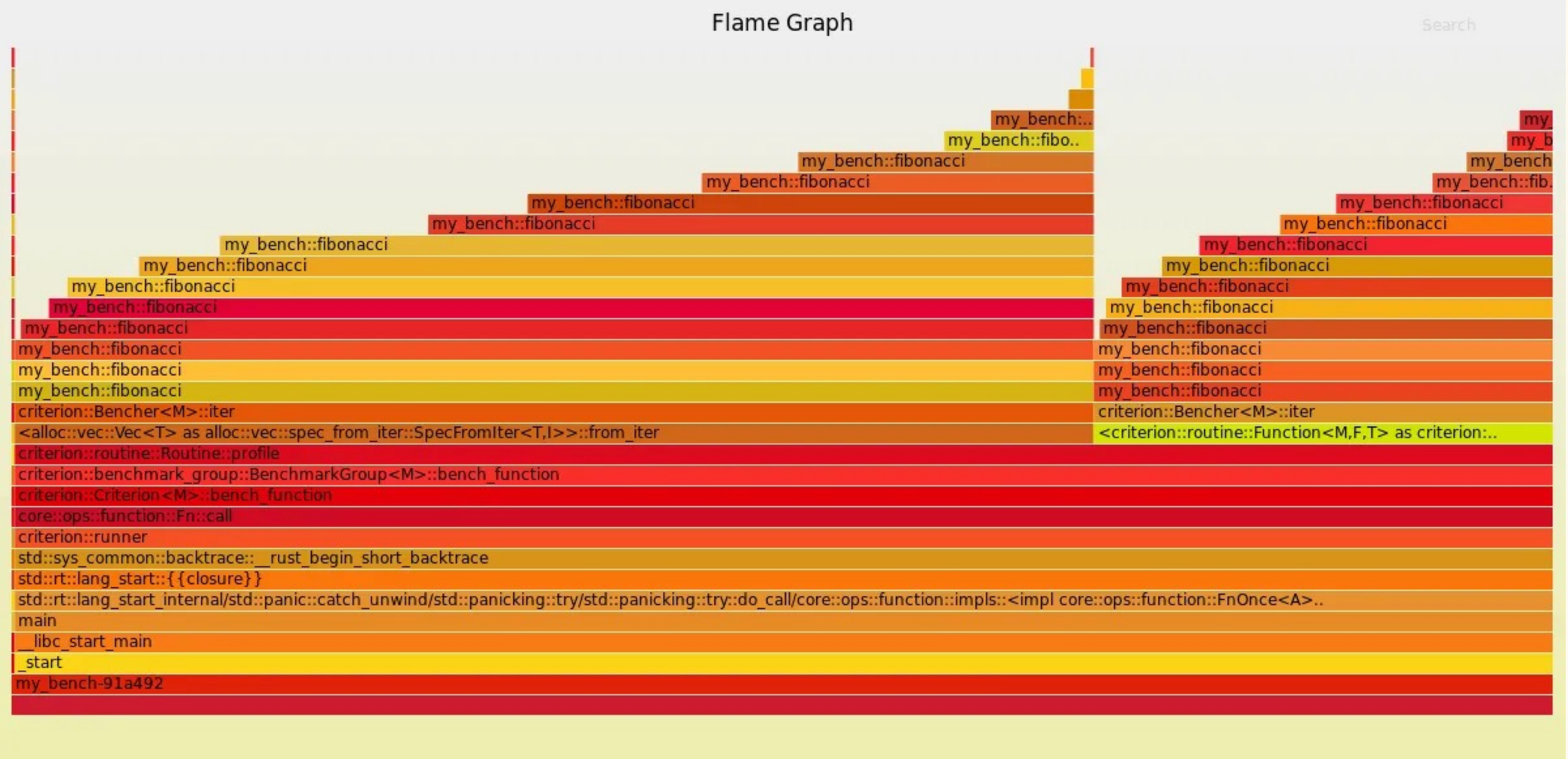
```rust
#[cfg(test)]
mod tests {
    use super::*;

    #[test]
    fn test_valid_index() {
        let result = get_item(1);
        assert_eq!(result, 20);
    }

    #[test]
    #[should_panic]
    fn test_invalid_index() {
        get_item(10);  // This should panic
    }

    #[test]
    #[should_panic(expected = "index out of bounds")]
    fn test_panic_message() {
        get_item(100);  // Panics with specific message
    }
}
```

Testing asynchronous

```rust
// Async function that can fail
async fn fetch_data(id: u32) -> Result<String, String> {
    if id == 0 {
        return Err("Invalid ID".to_string());
    }

    if id == 999 {
        return Err("Not found".to_string());
    }

    Ok(format!("Data for ID {}", id))
}
```

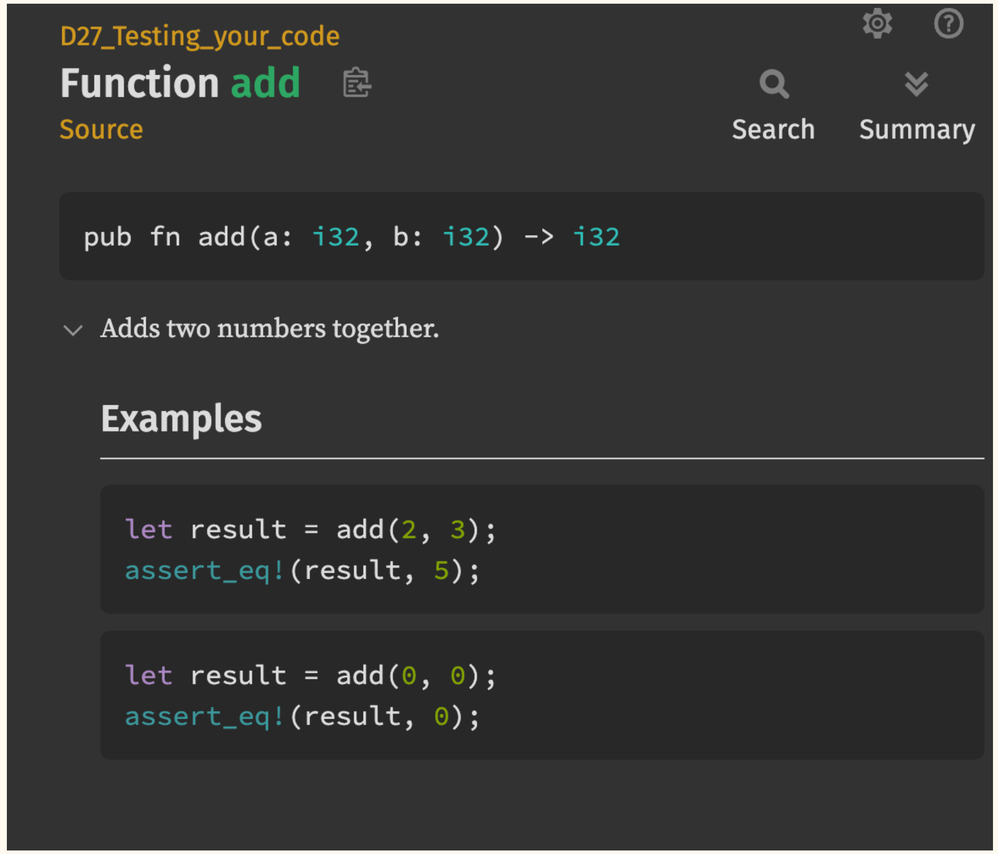
```rust
#[cfg(test)]
mod tests {
    use super::*;

    #[tokio::test]
    async fn test_fetch_success() {
        let result = fetch_data(1).await;
        assert_eq!(result, Ok("Data for ID 1".to_string()));
    }

    #[tokio::test]
    async fn test_fetch_invalid_id() {
        let result = fetch_data(0).await;
        assert!(result.is_err());
        assert_eq!(result, Err("Invalid ID".to_string()));
    }

    #[tokio::test]
    async fn test_fetch_not_found() {
        let result = fetch_data(999).await;
        assert_eq!(result, Err("Not found".to_string()));
    }
}
```

Benchmarking with Criterion



docs.rs & tests

```rust
/// Adds two numbers together.
///
/// # Examples
///
/// ```
/// # use D27_Testing_your_code::add;
/// let result = add(2, 3);
/// assert_eq!(result, 5);
/// ```
///
/// ```
/// # use D27_Testing_your_code::add;
/// let result = add(0, 0);
/// assert_eq!(result, 0);
/// ```
pub fn add(a: i32, b: i32) -> i32 {
    a + b
}
```



Let’s code!

Your turn!